

Unit 2 – Exploring Derivatives

THE CONSTANT RULE

If $f(x) = c$, c is a constant, then $f'(x) = 0$.

Proof:

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{c - c}{h} \\ &= \lim_{h \rightarrow 0} \frac{0}{h} \\ &= 0 \end{aligned}$$

Example: Determine the derivative for each of the following.

a. $f(x) = 0$

b. $f(x) = \sqrt{7}$

c. $y = -\frac{1}{2}$

d. $y = -4.67$

$$f'(x) = 0$$

$$f'(x) = 0$$

$$y' = 0$$

$$\frac{dy}{dx} = 0$$

THE POWER RULE

If $f(x) = x^n$, where $n \in \mathbb{R}$, then $f'(x) = nx^{n-1}$.

Proof:

Let $f(x) = x^n$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(x+h)^n - x^n}{h}$$

$$= \lim_{h \rightarrow 0} \frac{x^n + nx^{n-1}(h) + \frac{n(n-1)}{2}x^{n-2}(h^2) + \dots + (h)^n - x^n}{h}$$

Factor out the "h"

$$= \lim_{h \rightarrow 0} \frac{\cancel{x^n} + nx^{n-1} + \frac{n(n-1)}{2}x^{n-2}(h) + \dots + (h^{n-1})}{\cancel{h}}$$

Let "h" go to zero

$$= nx^{n-1}$$

Example: Determine the derivative for each of the following.

$$y = x^{\frac{1}{2}}$$

a. $f(x) = x$

$$f'(x) = x^0 = 1$$

b. $f(x) = x^5$

$$f'(x) = 5x^4$$

c. $y = \sqrt{x}$

$$y' = \frac{1}{2} x^{-\frac{1}{2}} = \frac{1}{2\sqrt{x}} \rightarrow x^{\frac{1}{2}}$$

d. $y = x^{-4}$

$$\frac{dy}{dx} = -4x^{-5} = \frac{-4}{x^5}$$

e. $f(x) = \frac{1}{\sqrt[3]{x}}$

$$f(x) = x^{-\frac{1}{3}} \\ f'(x) = -\frac{1}{3} x^{-\frac{4}{3}}$$

$$= \frac{-1}{3\sqrt[3]{x^4}} = \frac{-1}{3x^{4/3}}$$

Both ok!

f. $f(x) = \sqrt[5]{x}$

$$f'(x) = \frac{1}{5} x^{-\frac{4}{5}} = \frac{1}{5 \cdot \sqrt[5]{x^4}} = \frac{1}{5x^{4/5}}$$

THE CONSTANT MULTIPLE RULE

If $f(x) = c \cdot g(x)$, c is a constant, then $f'(x) = c \cdot g'(x)$.

Example: Determine the derivative for each of the following.

a. $f(x) = 4x^3$

$$f'(x) = 12x^2$$

b. $f(x) = -16x^{-4}$

$$\begin{aligned} f'(x) &= 64x^{-5} \\ &= \frac{64}{x^5} \end{aligned}$$

c. $y = -2x^{\frac{3}{2}}$

$$\begin{aligned} \frac{dy}{dx} &= -3x^{\frac{1}{2}} \\ &= -3\sqrt{x} \end{aligned}$$

d. $y = \sqrt{6}x^{-\frac{3}{5}}$

$$y' = \frac{-3\sqrt{6}}{5} x^{-\frac{8}{5}}$$

$$= \frac{-3\sqrt{6}}{5 \cdot \sqrt[5]{x^8}}$$

e. $y = 3x^{\sqrt{5}}$

$$\frac{dy}{dx} = 3\sqrt{5} x^{\sqrt{5}-1}$$

f. $y = \pi x^\pi$

$$y' = \pi^2 \cdot x^{\pi-1}$$

SUM & DIFFERENCE RULES

If both $p(x)$ and $q(x)$ are differentiable and $f(x) = p(x) \pm q(x)$, then $f'(x) = p'(x) \pm q'(x)$.

Example: Determine the derivative for each of the following.

a. $f(x) = \sqrt{\frac{2}{x}} + \sqrt{\frac{x}{3}}$

$$f(x) = \sqrt{2} x^{-\frac{1}{2}} + \frac{x^{\frac{1}{2}}}{\sqrt{3}}$$

$$f'(x) = -\frac{\sqrt{2}}{2} x^{-\frac{3}{2}} + \frac{x^{-\frac{1}{2}}}{2\sqrt{3}}$$

$$= \frac{-\sqrt{2}}{2\sqrt{x^3}} + \frac{1}{2\sqrt{3}x}$$

b. $y = -\frac{\sqrt{2}}{2} x^{-\frac{3}{2}} + \frac{x^{\frac{1}{2}}}{2\sqrt{3}}$

$$y' = \frac{3\sqrt{2}}{4} x^{-\frac{5}{2}} - \frac{x^{-\frac{3}{2}}}{4\sqrt{3}}$$

$$= \frac{3\sqrt{2}}{4\sqrt{x^5}} - \frac{1}{4\sqrt{3}x^3}$$

$$\nearrow x^{\frac{1}{2}}$$

c. $y = x^\pi - \pi x^3$

$$y' = \pi x^{\pi-1} - 3\pi x^2$$

$$y' = \pi (x^{\pi-1} - 3x^2)$$

d. $f(x) = 7x^4 - \sqrt{x} + 3x^{\frac{3}{2}} - 401$

$$f'(x) = 28x^3 - \frac{1}{2\sqrt{x}} + \frac{9}{2}\sqrt{x}$$

Example: Determine the equation of the tangent to the graph of $y = x^3 + 2x^2 - 4x + 1$ at $x = 4$.

$$\frac{dy}{dx} = 3x^2 + 4x - 4$$

$$y - y_1 = m(x - x_1)$$

$$m_t|_{x=4} = 48 + 16 - 4 = 60$$

$$y - 81 = 60(x - 4)$$

Example: Determine the values(s) of x so that the tangent to $f(x) = \frac{3}{\sqrt[3]{x}}$ is parallel to the line $x + 16y + 3 = 0$.

$$f(x) = 3 \cdot x^{-\frac{1}{3}}$$

$$x + 16y + 3 = 0$$

$$f'(x) = -x^{-\frac{4}{3}}$$

$$y = -\frac{1}{16}x - \frac{3}{16}$$

$$= -\frac{1}{3\sqrt[3]{x^4}}$$

$$\therefore m_t = -\frac{1}{16}$$

$$\therefore -\frac{1}{3\sqrt[3]{x^4}} = -\frac{1}{16}$$

$$\Rightarrow 3\sqrt[3]{x^4} = 16 \rightarrow x^{\frac{4}{3}} = 16 \rightarrow x = 16^{\frac{3}{4}}$$

$$\therefore x = \pm 8$$

Additional Questions

1. Determine the derivative of each of the following.

a) $f(x) = -\frac{3}{4}\sqrt[4]{x^5} - \frac{4}{3\sqrt{x^3}} + \pi^2 x^3 - \frac{7}{3}$

$$f(x) = -\frac{3}{4} x^{\frac{5}{4}} - \frac{4}{3} x^{-\frac{3}{2}} + \pi^2 x^3 - \frac{7}{3}$$

$$f'(x) = -\frac{15}{16} x^{\frac{1}{4}} + 2 x^{-\frac{5}{2}} + 3\pi^2 x^2$$

$$= -\frac{15\sqrt[4]{x}}{16} + \frac{2}{\sqrt{x^5}} + 3\pi^2 x^2$$

b) $g(x) = 4\sqrt[4]{x^3}(\pi\sqrt[5]{x} - 2^3\pi)$

$$g(x) = 4\pi x^{\frac{19}{20}} - 32\pi x^{\frac{3}{4}}$$

$$g'(x) = \frac{19}{5}\pi x^{-\frac{1}{20}} - 24\pi x^{-\frac{1}{4}}$$

$$= \frac{19\pi}{5\sqrt[20]{x}} - \frac{24\pi}{\sqrt[4]{x}}$$

c) $g(x) = \frac{3x^4 + 2\pi x^3 - 5\sqrt[3]{x}}{4x^2}$

$$g(x) = \frac{3x^2}{4} + \frac{\pi x}{2} - \frac{5}{4} x^{-\frac{5}{3}}$$

$$g'(x) = \frac{3x}{2} + \frac{\pi}{2} + \frac{25}{12} x^{-\frac{8}{3}}$$

$$= \frac{3x}{2} + \frac{25}{12\sqrt[3]{x^8}} + \frac{\pi}{2}$$